CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No. 09



By- CHANDAN SIR



01 HS

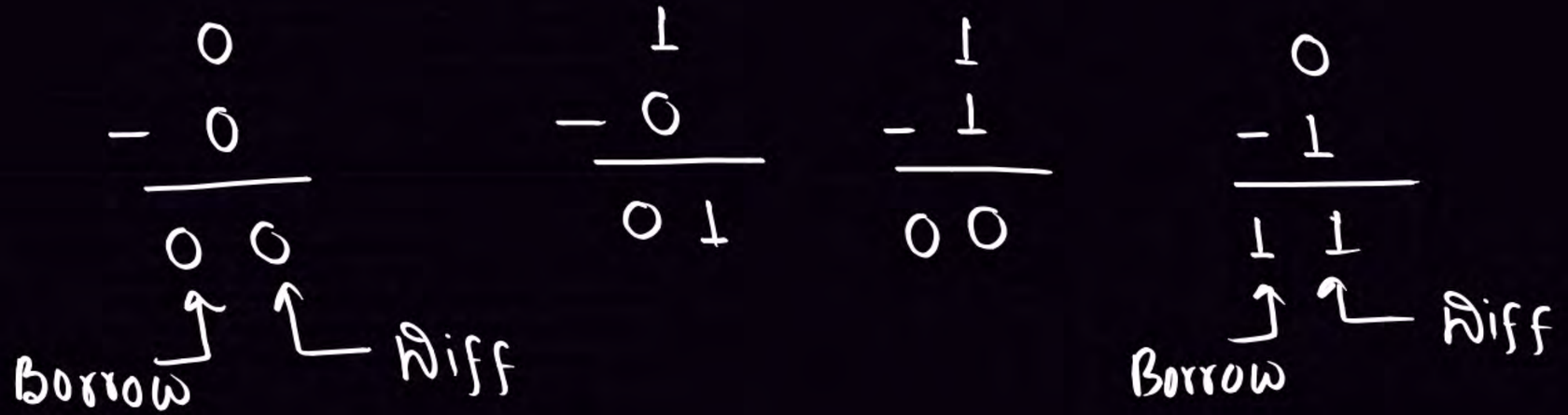
02 FS

03 Question Practice

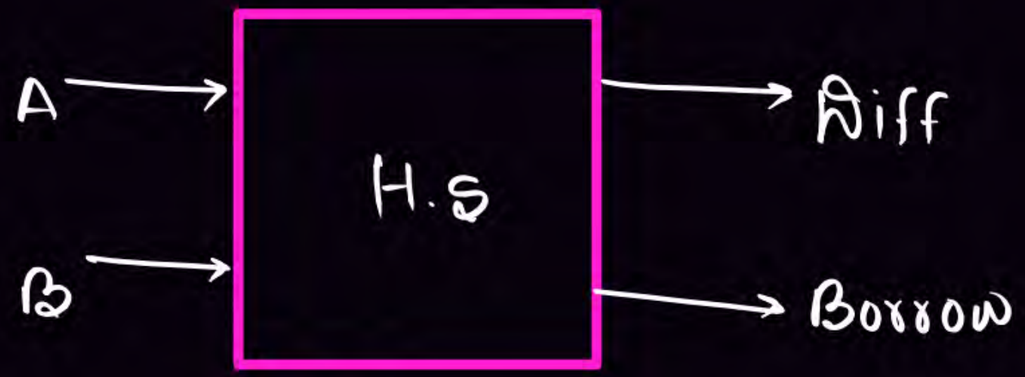
04 Discussion

HALF SUBTRACTOR :→

↳ Two bit subtractor



Step 1 :



Step 2.

A - B

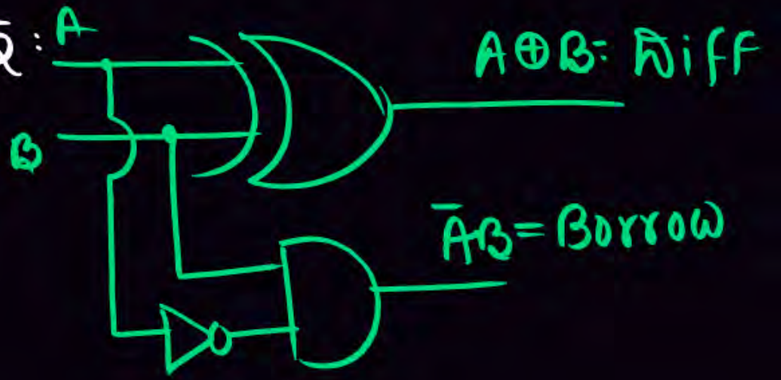
A	B	Diff	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

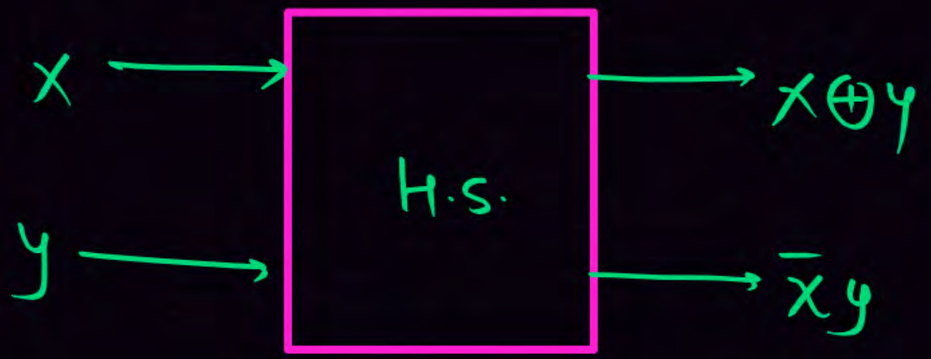
Step 3 : $Diff = A \oplus B$

$Borrow = \bar{A}B$

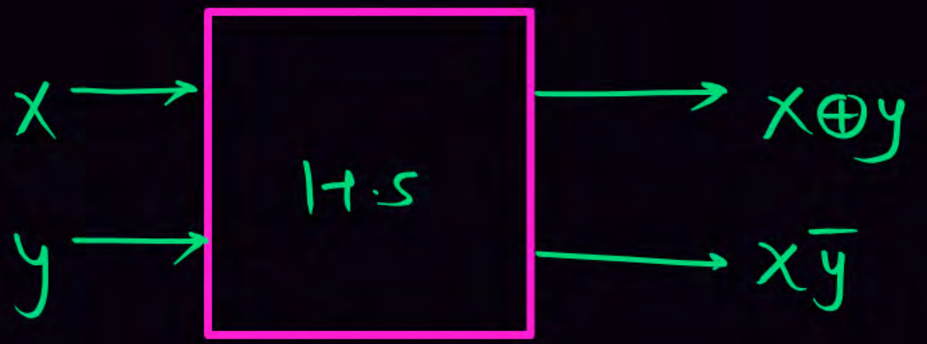
Step 4.

Step 5 :



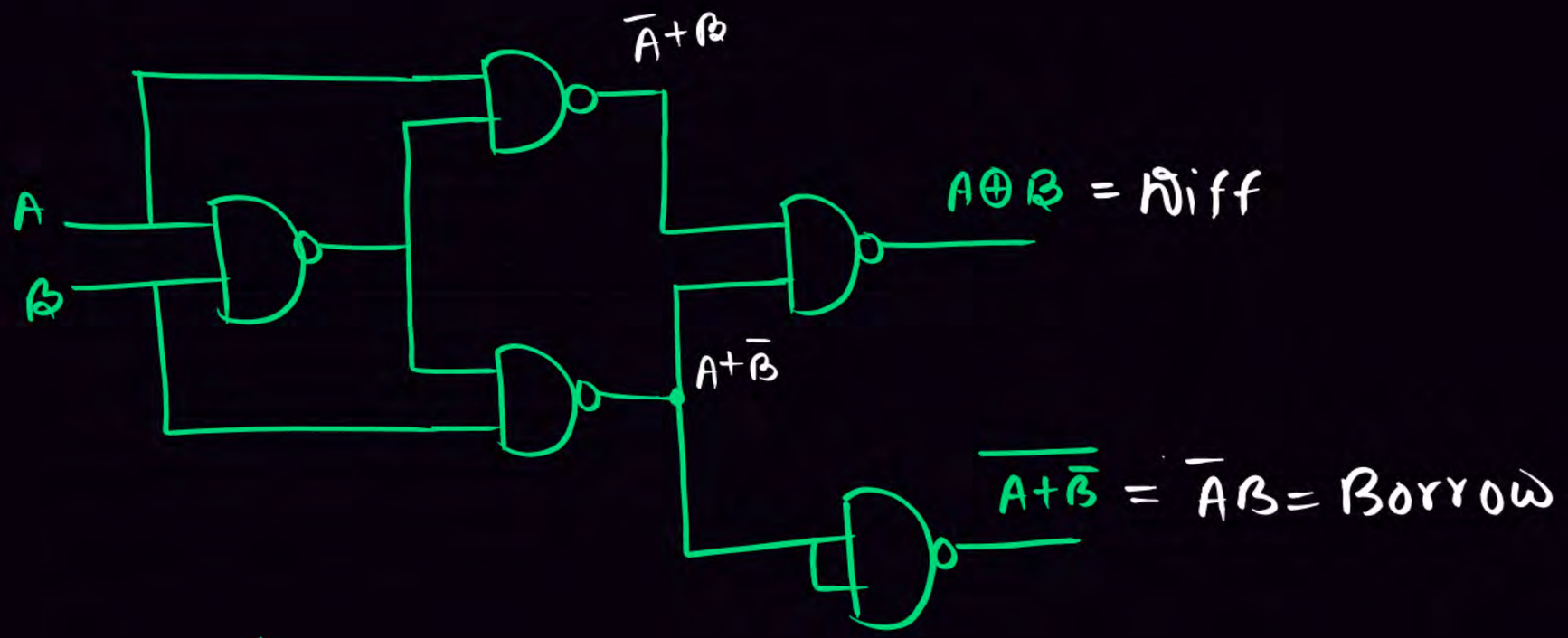


X-Y

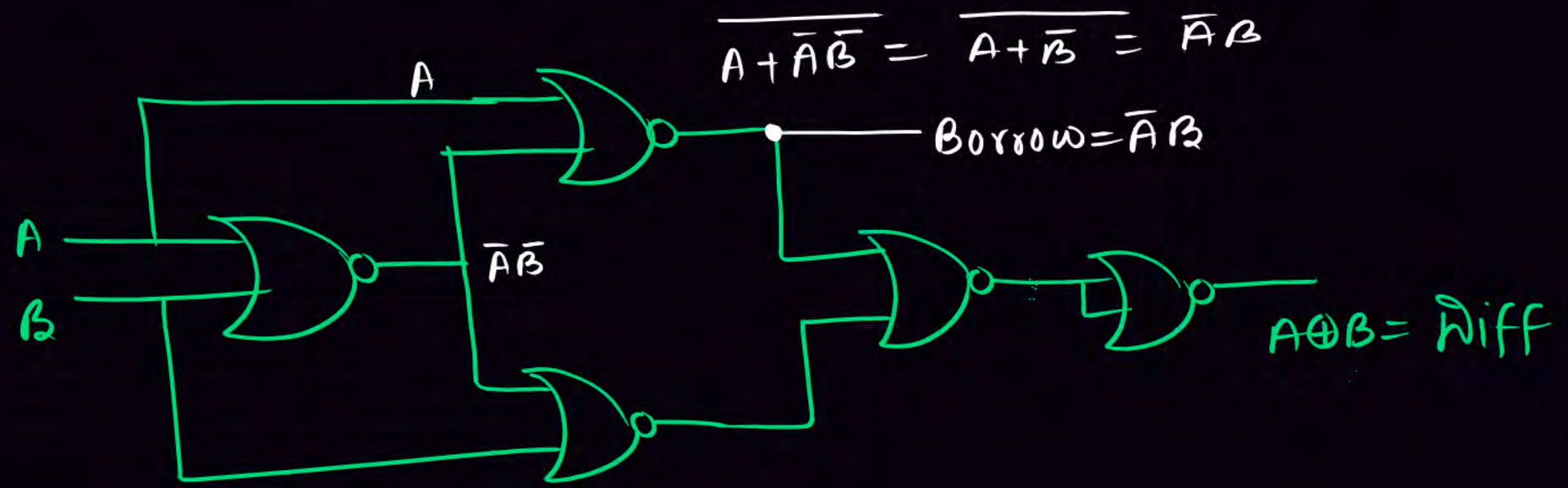


Y-X

H.S. by using NAND



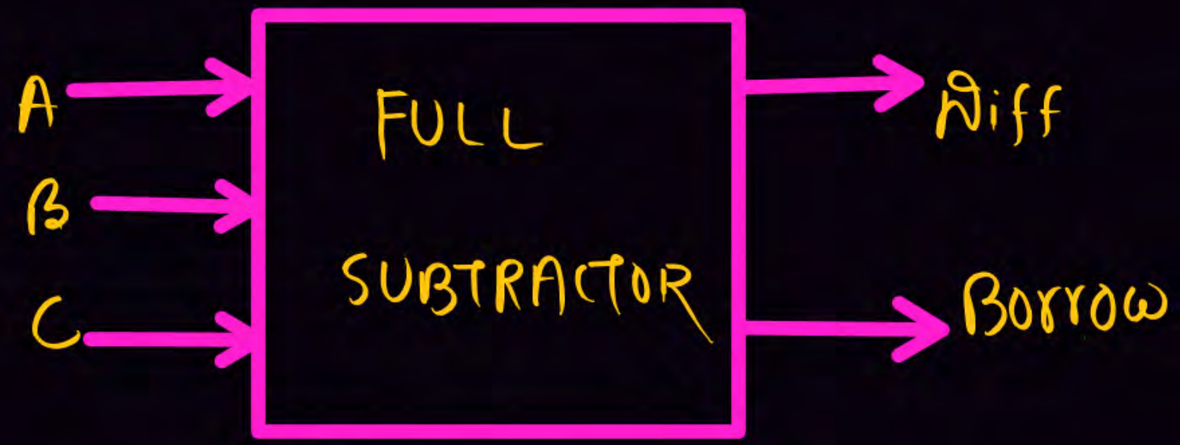
H.S. by using NOR



FULL SUBTRACTOR

→ Three bit subtractor

Step1:



Step 2:

$(A-B)-C$

A	B	C	Diff	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

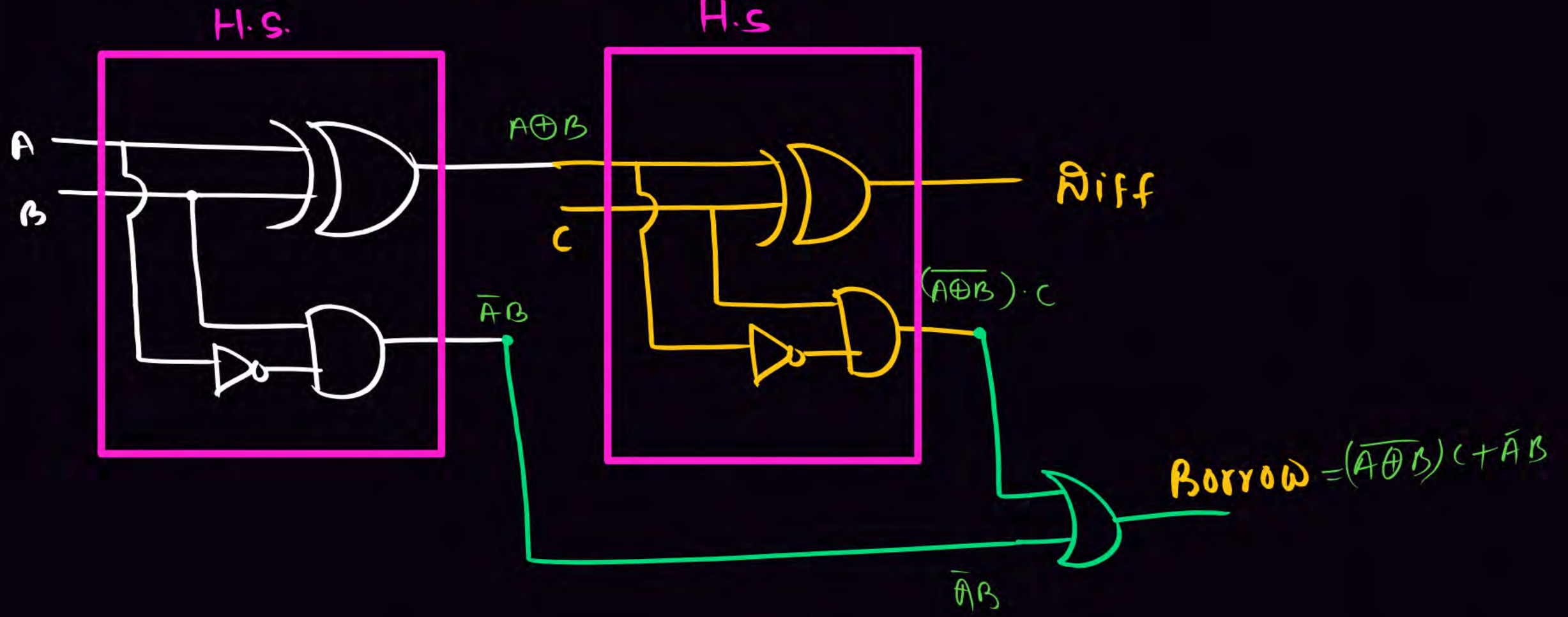
Step 3: & Step 4:

$$Diff = \Sigma m(1,2,4,7)$$
$$= A \oplus B \oplus C$$

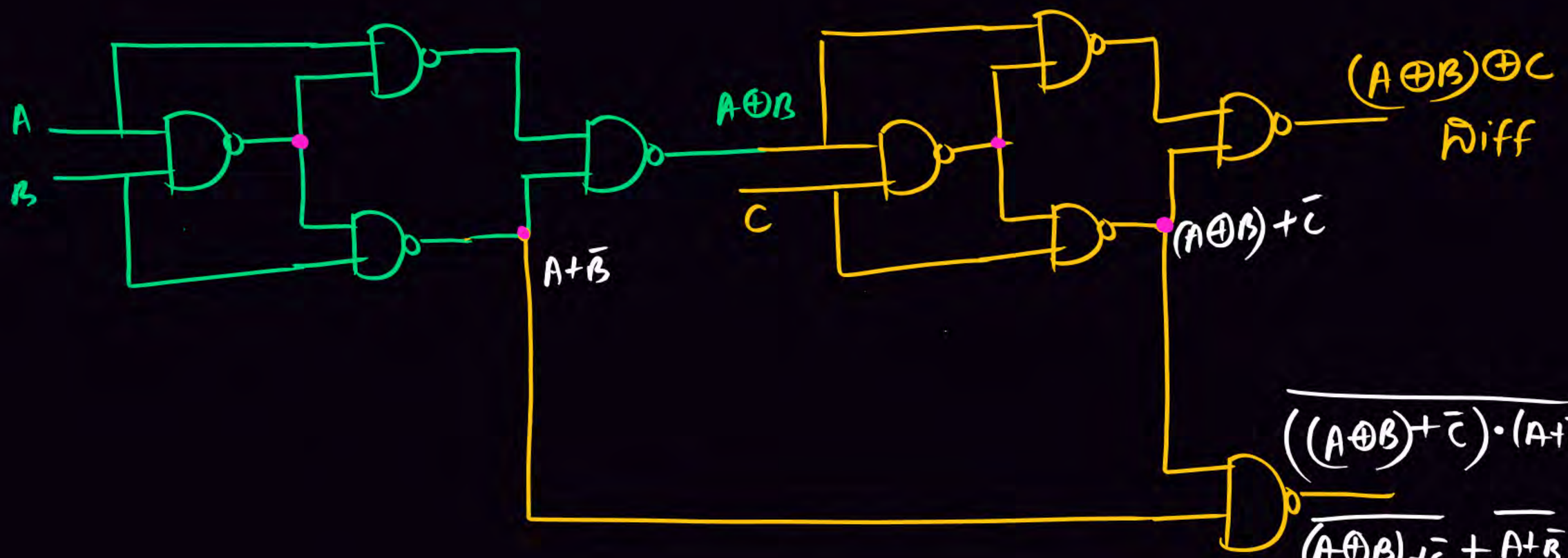
$$Borrow = \Sigma m(1,2,3,7) \quad \text{—①}$$
$$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC$$
$$= (\bar{A}\bar{B} + AB)C + \bar{A}B(\bar{C}+C)$$
$$= \overline{(A \oplus B)}C + \bar{A}B \quad \text{—②}$$
$$= \bar{A}B + BC + \bar{A}C \quad \text{—③}$$

Step5 :

1 Full subtractor = 2Hs + 1OR GATE



F.S. by NAND GATE

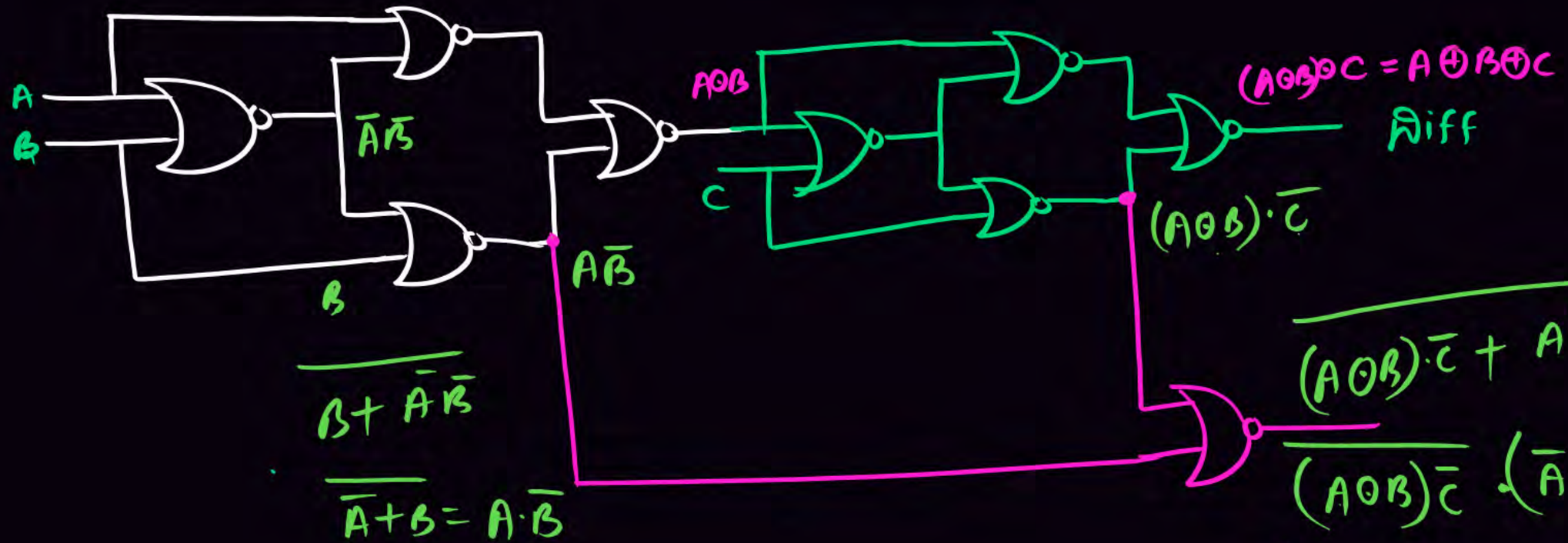


$$\overline{((A\oplus B)+\bar{C})\cdot(A+\bar{B})}$$

$$\overline{(A\oplus B)+\bar{C}}+\overline{A+\bar{B}}$$

$$\overline{A\oplus B}\cdot C+\bar{A}B=\text{Borr.}$$

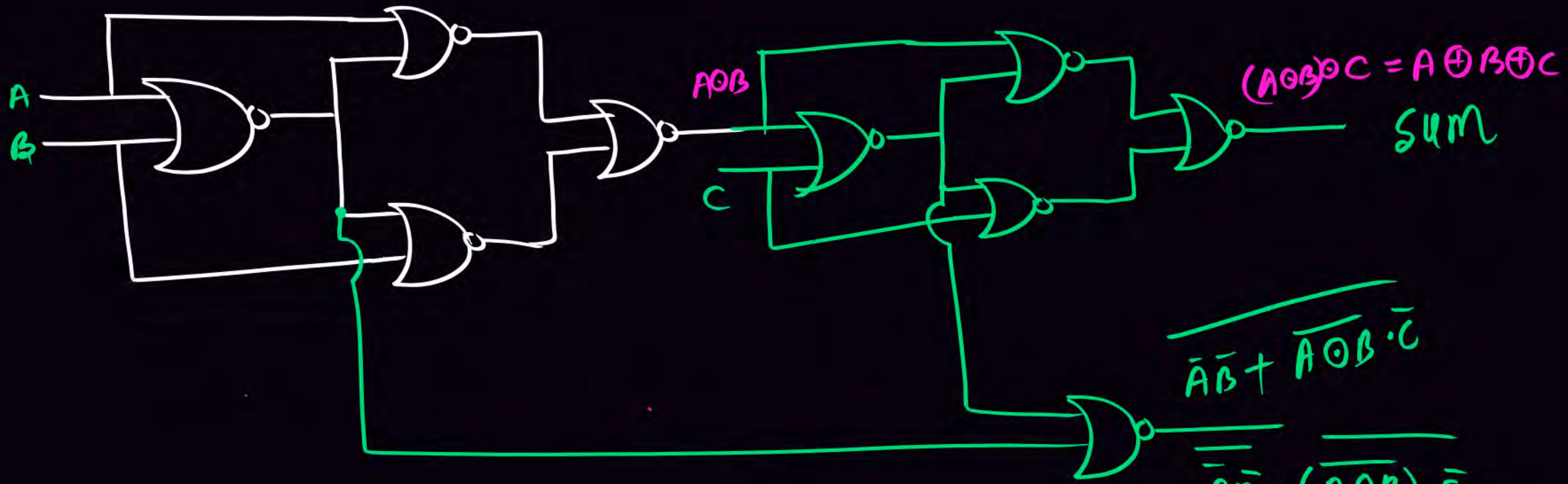
F.S. : by using NOR GATE :-



$\bar{A}B+\bar{A}C+BC$

Borrow

F.A :- by using NOR GATE :-

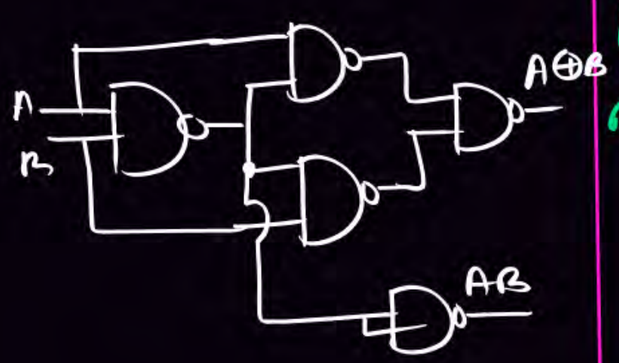


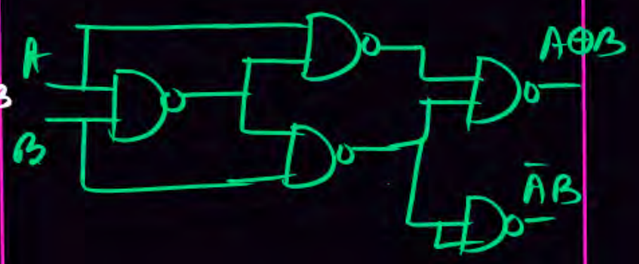
$(A+B)\{\bar{A}\bar{B}+AB+C\}$

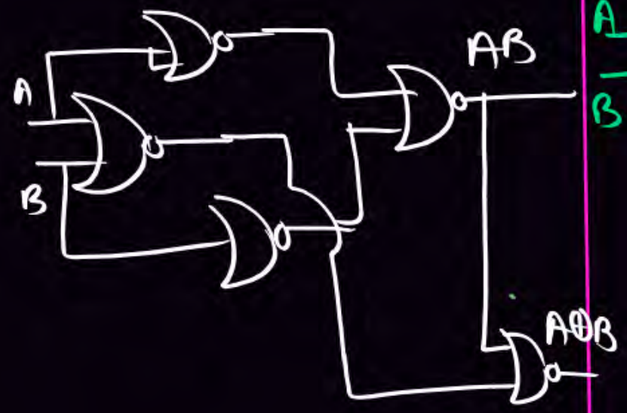
$(A+B)\{(A \odot B)+C\}$

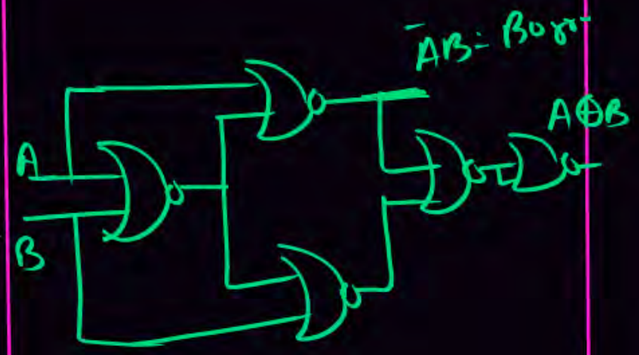
$AB+AC+AB+BC = AB+BC+AC = carry$

HA	H.S.	FA	F.S.
Sum = $A \oplus B$ Carry = AB	Diff = $A \oplus B$ Borrow = $\bar{A}B$	Sum = $\Sigma m(1,2,4,7)$ $= A \oplus B \oplus C$ Carry = $\Sigma m(3,5,6,7)$ $= (A \oplus B)C + AB$ $= AB + AC + BC$ NAND/NOR = 9 1 FA = 2 HA + 1 OR	Diff = $\Sigma m(1,2,4,7)$ $= A \oplus B \oplus C$ Borrow = $\Sigma m(1,2,3,7)$ $= (\overline{A \oplus B})C + \bar{A}B$ $= \bar{A}B + \bar{A}C + BC$ NAND/NOR = 9 1 F.S = 2 H.S + 1 OR

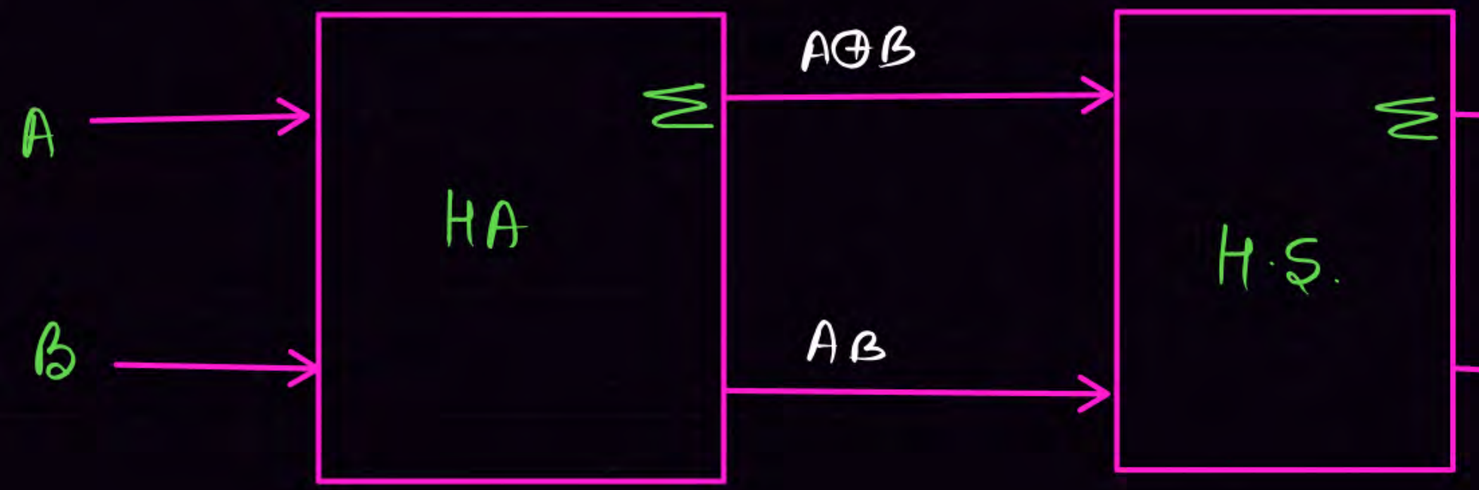








Q $y_0 = ?$ $y_1 = ?$



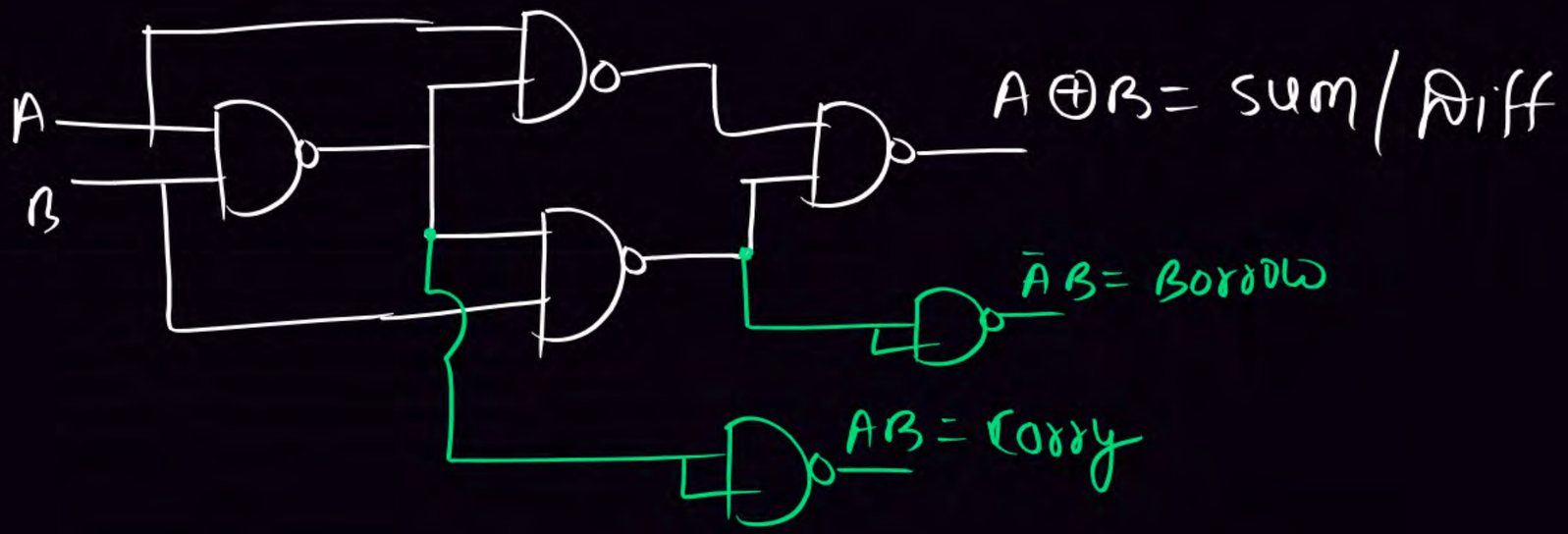
$y_0 = AB \oplus (A \oplus B)$

$= A + B$

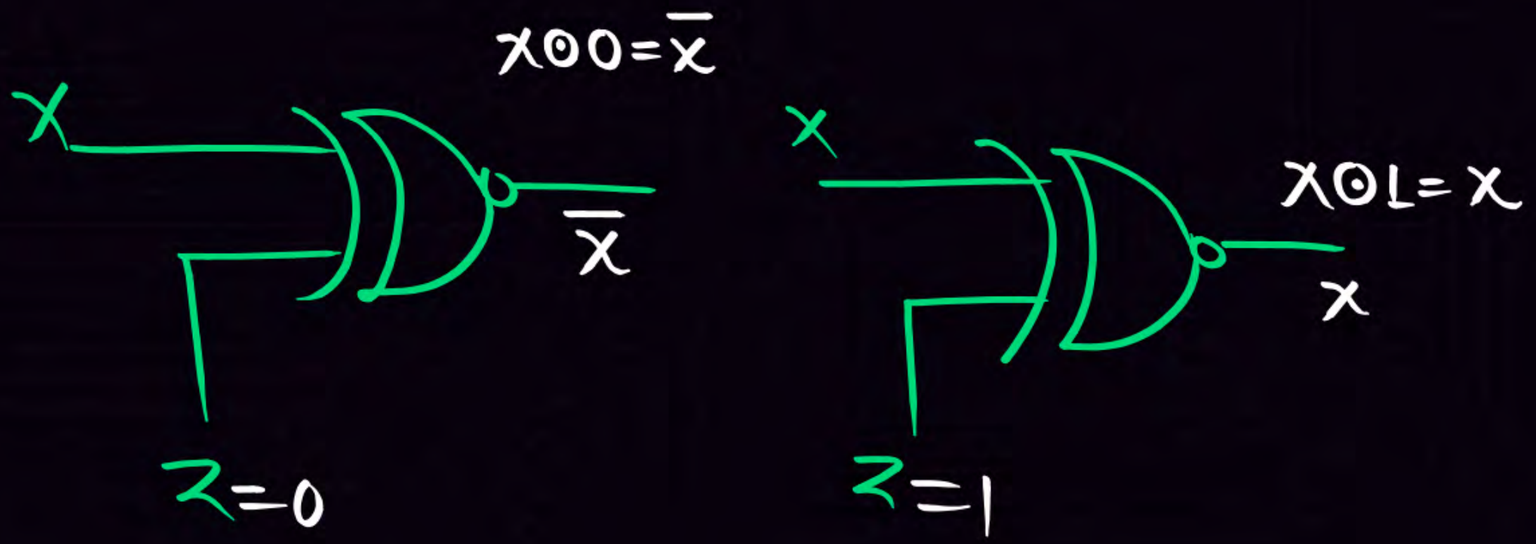
$y_1 = \overline{A \oplus B} \cdot AB$

$= (\bar{A}\bar{B} + AB)\, AB$

$= AB$



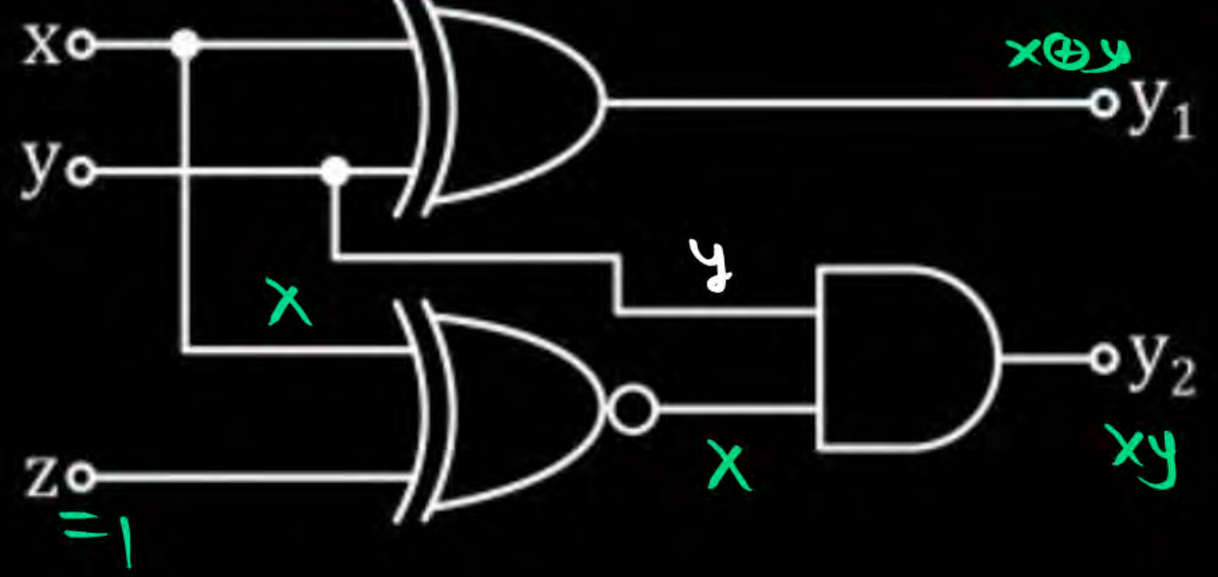
6 Ans


X00=X̄
X
X̄
Z=0
X
X01=X
X
Z=1

Question **(MCQ)**

#Q. The circuit shown below, is a controlled half adder/half subtractor. The inputs to half adder/half subtractor are x and y while z is a control. The outputs are y_1 and y_2.
Then the circuit behaves as

z = 0 H.S. x−y
z = 1 H.A x+y



(A) Half adder for z = 0

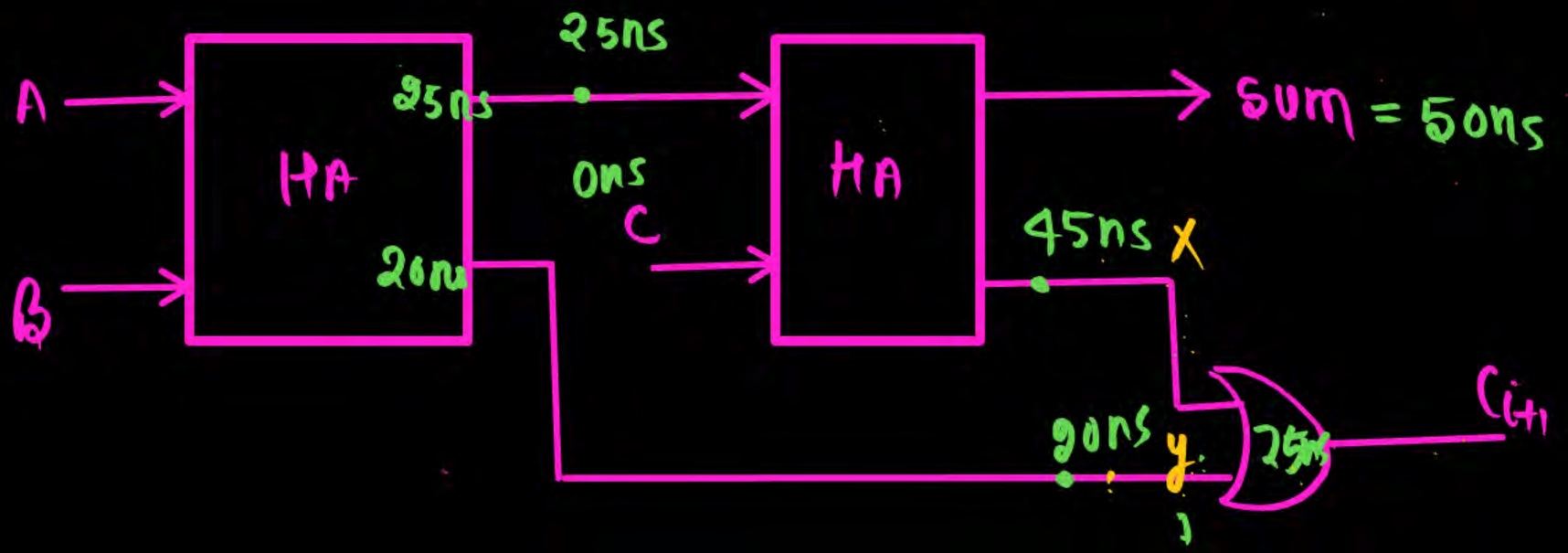
(B) Half subtractor for z = 1

(C) Half adder for z = 1 and half subtractor for z = 0

(D) Half adder regardless of weather z = 0 or z = 1 due to design defect

Question **(MCQ)**

A full adder is implemented with two half adders and one OR gate. OR gate is used to derive the final carry function of full adder. In each half adder, T_{sum} = 25 ns and T_{carry} = 20 ns and T_{OR} = 25 ns. The max minimum time required to derive both the sum and carry function of a full adder after applying the inputs is ~~50~~/70 ns



case(1) Maximum (y=0)

T_C = 45ns + 25ns

= 70ns ✓

T_S = 50ns

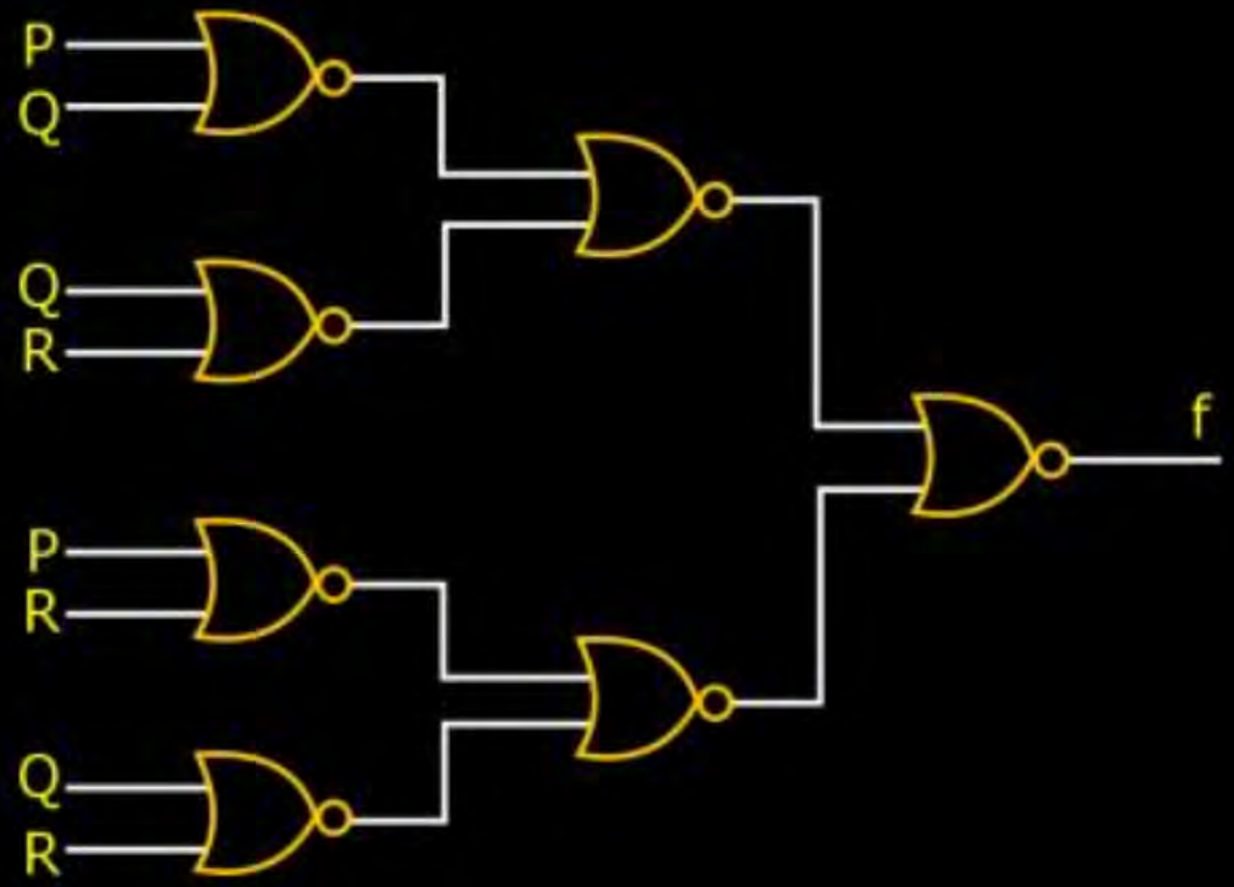
case(2) Minimum (y=1

T_C = 20 + 25 = 45ns

T_S = 50ns

Question **(MCQ)**

What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below?



A $\overline{Q+R}$

B $\overline{P+Q}$

C $\overline{P+R}$

D $\overline{P+Q+R}$

Thank you DOSTO